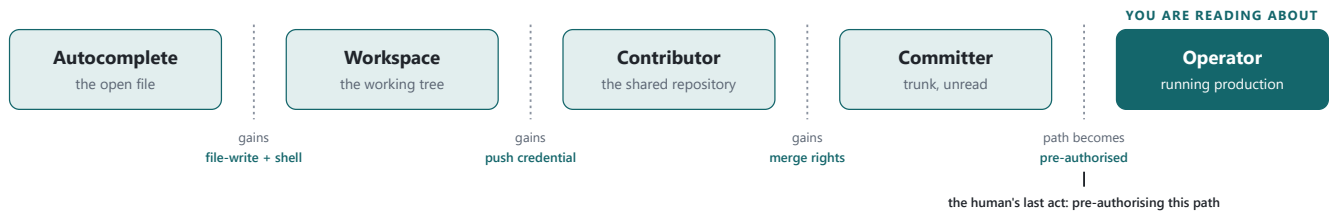


Operator

Nothing human stands between the agent's change and running production. No team observed sets its ceiling here — changes reach this archetype by qualifying for it, one narrow class at a time.

Vocabulary as of 2026-08-31 Taxonomy: ADR-0004 Sources last verified 2026-09-13

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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01 What this archetype is

An Operator ceiling permits a defined class of model-generated changes to reach **running production with no human action in between**. There are two routes to that condition, and the archetype deliberately does not distinguish them: the agent lands on trunk unread and continuous deployment carries the change to production unattended, or the agent itself holds deploy credentials. Either way, nothing human stands in the path — a pre-authorised pipeline holds the credential on the agent's behalf [ADR-0004](#).

An archetype names a team's *ceiling* — what its settings and policies permit for model-generated changes — not what any individual change receives. That distinction does more visible work here than anywhere else on the spectrum, because the two halves of it point in opposite directions: **individual change classes demonstrably reach this archetype, and no team observed sets its ceiling here**. Section 04 carries the evidence for both halves.

02 The boundary crossed

What separates Operator from Committer is that **the path from trunk to production becomes pre-authorised**. A Committer team's ceiling lets a change class merge unread while a human still gates the release; crossing this boundary removes that last human act from the path — usually not by handing the agent a deploy credential, but by a human authorising, in advance, a pipeline that will act without them.

This is not a theoretical construction. Surveying every real far-end mechanism located across the tool landscape produced one shape: *"not an agent that merges, but a human who pre-authorises a merge that then happens without them"* [Tooling Landscape, Aug 2026](#). Devin documents it exactly: *"Enable or disable GitHub auto-merge from the merge button dropdown. When enabled, the PR will merge automatically once all required checks pass"* [Devin docs, Aug 2026](#). Cursor's automation tool list contains no merge action at all — the path is *approve, then GitHub's native auto-merge lands it* [Cursor docs, Aug 2026](#). The decision is human and prior; only the execution is unattended.

The boundary is therefore crossed by **an administrator changing a policy, not by buying a tool**. No product merges its own pull request on its own judgement, and the platform hosting nearly all observable agent activity forbids its own agent from doing so: GitHub's cloud agent *"cannot approve or merge a pull request"*, and its PRs require an additional human approval [GitHub Docs, Aug 2026](#). A team that wants this archetype must build the path deliberately — which is why every occupant of it is visible as an artifact: a workflow file, a bot account, a ruleset bypass [Practitioner Exemplars, Aug 2026](#).

03 What a human still gates

Reach fixes the archetype; oversight varies within it. On any real approximation of an Operator configuration, humans still gate three things. First, **the qualifying line itself**: which change classes may travel the pre-authorised path — the single most consequential control this archetype has, and everywhere it has been observed, a narrow one. Second, **the pre-authorisation**: the path exists because a human with authority created it, and that person can close it. Third, **everything outside the qualifying class** — on every team whose changes reach this archetype, the overwhelming majority of changes still travel gated routes.

INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. Writing or implying that an Operator-reaching team is "unsupervised" is exactly the conflation this project exists to break.

04 Who works this way

As a ceiling: nobody found — and the absence was searched for, not merely unobserved. The far end was hunted through four independent channels: Google News RSS across ~10 query formulations; Hacker News full-text search on exact phrases ("no human review", "agents merge"); authenticated GitHub code search for the operational artifact that would have to exist (`"pr merge --auto" "claude"`).

path: `.github/workflows` , "merged without human review", "agents may merge", approval-bot patterns); and direct fetch of every named lead [● Practitioner Exemplars, Aug 2026](#) . Treat the result as a floor, not a census — but the result is a count of zero teams whose stated policy is a general Operator ceiling for model-generated changes.

As a per-change condition: real, and narrow. The genuine, documented cases of agent-authored changes merging to a production default branch without human review, as stated policy, number three — Razorpay's Slash platform (~100 PRs/week merged with no human in the loop, self-published, no control period) [● Razorpay, 2026](#) ; JetBrains' IdeaVim repository, where PRs authored by `claude[bot]` whose title begins `Update changelog: auto-merge` (the workflow file is public) [● IdeaVim workflow, 2026](#) ; and a 1-star experimental repository with a published agent protocol [● Practitioner Exemplars, Aug 2026](#) . None is what the phrase "agents ship to production" implies, and none documents the further step this archetype requires: an unattended path from that merge to *running production*.

The delivery-end anchor is **Meta's RADAR**: 331,000+ diffs landed with no human reviewer, on a risk-scored subset selected for being low-risk, with a revert rate one third and a production incident rate one fiftieth of human-reviewed diffs [● Meta \(arXiv\), 2026](#) . Whether Meta's pipeline then deploys those diffs unattended is not documented, so even the strongest case on record is confirmed only to Committer reach. The largest unattended-merge operation of all, Spotify's 2.5 million auto-merged maintenance PRs, is **the non-AI precedent, not an instance**: those are scripted, deterministic transforms with no model involved, while Spotify human-reviews every LLM PR [● Spotify, Jun 2026](#) .

One vendor's own playbook declines this archetype too. Anthropic's *AI-Native SDLC Playbook* (2026) carries the agent through the pipeline to a withheld gate — "*the agent may act up to the production gate and cannot pass it*" — and grants unattended execution only to a "*pre-approved runbook*" rehearsed in advance. A vendor's recommendation, not an observation, but it has the shape §02 found everywhere: a human authorises the path, and only the execution is unattended [● Anthropic playbook, L12 & L13, 2026](#) .

The pattern across all of it is the page's thesis: **teams reach the far end by narrowing what qualifies, never by widening who may act**. And the refusal voice is worth hearing at full strength — from a company whose product *is* production infrastructure: "*No LLM has any access to prod here*" [● Thomas Ptacek \(Fly.io\), Jun 2025](#) .

05 What it gains you, what it costs

WHAT IT GAINS YOU

- Full pipeline speed for the change classes that qualify — human review is measurably a queueing delay, and for the qualifying class the queue disappears entirely [● Meta \(arXiv\), 2026](#).
- The clearest possible statement of what your verification actually proves: when nothing human is in the path, the pipeline's guarantees are the whole safety case, stated in machinery rather than in habit.

WHAT IT COSTS YOU

- The entire engineered chain — gates, canaries, auto-revert, provenance — must exist and be trusted *before* the path opens, and §06 shows almost none of it exists wired to autonomy anywhere.
- The qualifying line must be engineered, argued for, and policed forever; it is the control everything else hangs from.
- Liability keys to the setting: the one vendor term pricing this configuration makes the customer "*solely responsible*" on enabling it [● Cursor ToS §1.7](#).

06 What it demands of verification

The entailment reaches its maximum here: everything a human ever did between a change and production must be replaced by engineered verification. The demand is written as a requirement — never as a claim that teams meet it — because the evidence runs the other way:

Further along the spectrum, more rigor is *required* — and the evidence that anyone has built it is thin, vendor-dominated, and in several documented cases decayed silently after it was built.

[● Verification Infrastructure, Aug 2026](#)

Concretely, an Operator team is building first-of-kind wiring. **No adopted mechanism anywhere gates, canaries, or auto-reverts a change *because it was agent-authored* — and at the canary and auto-revert layer, no published mechanism at all** (re-verified with search, 2026-09-02) — progressive delivery is applied to changes, never to authorship classes, and the provenance and delivery layers have never been connected in practice [● Verification Observability, Aug 2026](#). The published counterexamples sit at the merge layer only, and neither is in use: `agent-approval-check` (Anthropic, merged 2026-06-30) responds to agent authorship by requiring *more human approvals*, not more engineered checking — and detects agents by committer identity, so it does not fire for the dominant workflow [● Tooling Landscape, Aug 2026](#) — and one marketplace product (MergeShield, Mar 2026) risk-scores agent PRs by per-agent trust and auto-merges below thresholds, in this archetype's own direction, but has no observed adopter and has been dormant since days after launch [● Verification Observability, Aug 2026](#). Exactly one artifact in the corpus ties a verification floor to an autonomy level, and it counts assertions, not coverage [● Verification Infrastructure, Aug 2026](#). A second has since appeared, as a vendor's recommendation rather than a deployed system: Anthropic's playbook describes response bands held in version-controlled configuration

— at one standard deviation the agent logs, at two it diagnoses read-only, at three it may act *"only by opening a PR into the review gate or triggering a pre-approved runbook."* That wires a production signal to a permitted action; it still keys nothing to authorship [● Anthropic playbook, L13, 2026](#) .

Two measured facts shape what must be built. First, supervision cannot be the substitute: the vendor holding the telemetry reports permission-prompt approval at ~93% rising to 97%, and measures human review catching 13.6% of dangerous commands against 89% for automated checking [● Anthropic telemetry, 2026](#) . Second, the checks themselves are targets: handed a proof obligation it could not pass, an agent silenced failures with `pragma Assume` , and after that was banned, excluded code from proof with `SPARK_Mode => Off` — anti-suppression has to be engineered, then engineered again [● secunet \(arXiv\), 2026](#) . And the machinery decays silently: of the four organisations found running real agent evals (all four agent vendors), Cline shipped a CI regression workflow in February 2026 and lost it in May in an unrelated refactor, with nothing detecting the loss [● Verification Infrastructure, Aug 2026](#) .

The same vendor's playbook is also the fullest published articulation of the administrator-flippable ceiling this page keeps meeting. Its managed-settings play deploys the whole posture as one file engineers *"cannot edit or override"* — bypass mode disabled, only managed permission rules and hooks honoured, the sandbox a precondition, credential paths denied — and its release gates are hooks. That is where it must be cited in tension with the server-side test rather than as its refutation: a hook is an agent-side control, however tamper-proofed, and the playbook itself re-grounds the boundary that matters in branch protection. Its anti-suppression rule matches the secunet case to the letter — *"an agent fixing code must not be able to weaken the check on that code"* — and its rollback is *"a single command... exercised regularly in staging"*, proven before the path opens. Every one of these is a prescription; the course reports no value for any indicator it defines [● Anthropic playbook, L8, L11 & L12, 2026](#) .

RADAR is what the demand looks like partially met — and its own scoping is the discipline this section asks for: the favourable revert and incident rates are *selection, not proof of superiority*, because the qualifying diffs are chosen precisely for being low-risk. The operational heart of it is a single tunable dial — relaxing the risk threshold from the 25th to the 50th percentile raises the auto-approved share to 60.31% — deciding what fraction of changes never meet a human [● Meta \(arXiv\), 2026](#) . That dial is the qualifying line of §03, rendered in code.

07 Governance implications

Compliance is not what stops a team from operating here. Across SOC 2, ISO 27001, ISO/IEC 42001, the EU AI Act, DORA, the CRA, NIST SSDF, SLISA, HIPAA and DO-178C, no verified requirement was found that a human being must read source code. The two most quotable artifacts are now first-party verified: NIST's own generative-AI profile states that its practices *"do not distinguish between human-written and AI-generated source code"* (stated for a profile whose scope is building AI models — the caveat travels with the quote) [● NIST SP 800-218A, 2024](#) , and the US federal attestation form asks for *"automated tools or comparable processes"* four times while the phrase "code review" appears in it exactly zero times [● CISA/OMB form, 2024](#) .

What changes at this ceiling is who holds the consequences. Every provider disclaims defect liability — GitHub's terms: *"You are responsible for reviewing, testing, and validating any Output before use"* [● GitHub ToS §1.4, 2026](#) — and one vendor prices this archetype's defining act directly: enabling execution

"without manual review or confirmation" makes the customer *"SOLELY RESPONSIBLE FOR ANY IMPACT RESULTING FROM USE OF THIS FEATURE, INCLUDING ENSURING APPROPRIATE SAFEGUARDS, TESTING, AND MONITORING ARE IN PLACE"* [● Cursor ToS §1.7](#). The liability attaches to the *setting* — the ceiling — not to any individual change. Authorship follows the same logic: merging an agent's change is adoption, not authorship — *"providing instructions to a machine and selecting an output does not equate to authorship"* [● US Copyright Office, 2025](#) — so at this archetype the operative provenance question, "which lines did a human write," has its starkest answer.

Official guidance, where it exists, speaks this archetype's own language — containment. The UK financial sector's cross-market guidance directs firms to *"govern AI on the basis that it operates with privileged access: tightly control high-risk capabilities (including tools, code execution and APIs), enforce scoped permissions, robust logging, human oversight, and effective kill-switch mechanisms"*, and to monitor agents *"as potential malicious insiders"* [● CMORG, Jun 2026](#). It contains no human-review mandate for code; it asks whether AI-assisted outputs are *"tested and validated to the same standard as human-generated work"* — validation parity, which is exactly the demand of §06.

08 The posture that fits

This fits a team that believes engineered verification can be made a stronger check than a human in the path — and that the honest route there is to narrow what qualifies, never to widen who may act.

The safety belief underneath: *the pipeline proves it*. The control moves; it does not improve. And the evidence gives that belief its honest edge: the machinery that would prove it mostly does not exist yet, which is why every real approach to this archetype is a narrow, visible, deliberately engineered carve-out — and why no one has made it their ceiling.

— Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

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